

GAHWA: A Computational Lexicon for Emirati Arabic

Saeed Alzaabi¹

Salam Khalifa^{1,3}

Mostafa Saeed^{1,2}

Shahd Dibas²

Hanan Aldarmaki²

Nizar Habash^{1,2}

¹Computational Approaches to Modeling Language Lab, New York University Abu Dhabi

²Mohamed Bin Zayed University of Artificial Intelligence ³Stony Brook University

{saeed.alzaabi, salam.khalifa, mostafa.saeed, nizar.habash}@nyu.edu

{shahd.dibas, hanan.alarmaki}@mbzuai.ac.ae

Abstract

Emirati Arabic remains one of the least-resourced Arabic dialects, with lexical information scattered across corpora, morphological databases, and dictionaries developed under incompatible annotation schemes. We present Gahwa, a machine-readable lexicon constructed by extracting, enriching, and standardizing evidence from an annotated Emirati corpus, a Gulf Arabic verb database, and a Gulf Arabic Wiktionary, unifying their input into one lemma-POS keyed scheme. We evaluate Gahwa through automatic coverage analysis of three Emirati datasets, *GumarAE*, *Mixat*, and the *Ramsa Corpus*, against existing Arabic morphological resources, and through analysis of out-of-vocabulary (OOV) forms. Results show that Gahwa has high word coverage across the three datasets, achieving 89.1-94.2% effective diacritized lemma-POS token coverage, while the remaining OOV cases primarily reflect lexical gaps, orthographic variations, fused expressions, and morphological gaps.

1 Introduction

Computational resources for Arabic dialects remain unevenly distributed. Emirati Arabic is represented in several written and spoken datasets, but its lexical evidence is fragmented across corpora, analyzer-linked databases, and dictionary material. These resources were developed for different purposes, encode different lexical units, and are not all distributed under the same access model. Emirati material is also frequently grouped under a broad Gulf label. In practice, researchers who wish to develop Emirati NLP tools must first reconstruct and standardize evidence from separate resources.

We present Gahwa, a computational lexicon that brings three existing streams of Emirati and Gulf lexical evidence into a common machine-readable design. The resource separates an expanded view

of entry-level forms from a consolidated lexical view. The expanded view retains the form and analysis supplied by each source, while the consolidated view groups compatible records by lemma, part of speech, and English gloss. We additionally enrich fields that are absent from individual sources and retain the relation between each entry-level record and its consolidated lexical representation.

This form of integration is particularly useful in a low-resource setting. When annotated evidence is limited, the value of an individual source lies not only in its size but also in the information it contributes: corpus attestation, verbal paradigms, lexical categories, roots, glosses, or frequency evidence. Standardizing these sources avoids repeated reconstruction and establishes a shared lexical foundation for future Emirati Arabic NLP work.

Finally, we evaluate Gahwa through Emirati Arabic in-domain coverage analyses on *GumarAE*, *Mixat*, and the *Ramsa Corpus* using a number of existing Arabic morphological analyzers. Since no independently curated Emirati lexical gold standard is currently available, our evaluation focuses on analyzer coverage, Gahwa’s coverage of returned lemma-POS analyses, the impact of repeated-letter normalization, and an analysis of out-of-vocabulary (OOV) forms. Together, these analyses characterize both the current lexical coverage of Gahwa and the principal sources of remaining gaps.

We make Gahwa publicly available to support research on Arabic NLP.¹

The rest of the paper is structured as follows. We present related work in Section 2. This is followed by Gahwa design and construction details in Section 3. We evaluate the coverage of Gahwa in Section 4.

¹<https://github.com/CAMeL-Lab/Gahwa>

Resource	Type	Size	License	Reference
Gumar Corpus	Gulf Arabic Text Corpus	110M words	CC BY 3.0	Khalifa et al., 2016
Gumar AE Corpus	Emirati Gumar Subset	12M words	CC BY 3.0	Khalifa et al., 2016
EMALAC	Child-language Speech Corpus	78K words	Restricted DUA	Ntelitheos and Idrissi, 2017
Gulf Verb Lexicon	Verb Paradigms	2.6K lemmas	CC BY 3.0	Khalifa et al., 2017
Emirati <i>Gumar Ann</i>	Annotated Emirati Gumar Subset	200K words	CC BY 3.0	Khalifa et al., 2018
Gulf Wiktionary	Wiktionary	525 records	CC BY-SA 4.0	Krumbiegel, 2023
DiaLEX-UA	Emirati Full Form Lexicon	29K lemmas	ELRA END USER	Halpern, 2023
MIXAT	Emirati-English Speech Corpus	15 hours	CC BY-NC-SA 4.0	Al Ali and Aldarmaki, 2024
Traditional Emirati	Speech Corpus	4 hours	–	Alblooki et al., 2025
RAMSA	Speech Corpus	41 hours	Restricted DUA	Al-Sabbagh, 2026

Table 1: Published resources explicitly targeting Emirati Arabic or containing a dedicated Emirati component. Reported units differ and should not be interpreted as directly comparable measures of lexical coverage.

2 Related Work

Arabic Dialect Lexicons. Examples of machine-readable dialectal lexicons include the Egyptian Colloquial Arabic Lexicon and the lexical database underlying CALIMA EGY (Kilany et al., 2002; Habash et al., 2012); THARWA, which links Egyptian and Levantine Arabic with MSA and English (Diab et al., 2014); the 25-city-aligned MADAR Lexicon (Bouamor et al., 2018); the Iraqi Arabic Morphological Lexicon and related Iraqi and Moroccan lexicon-development work (Graff et al., 2006; Graff and Maamouri, 2009); MAKNUUNE, a richly structured Palestinian Arabic lexicon (Dibas et al., 2022). Qabas further links a large Arabic lexical inventory to many lexicons and morphologically annotated corpora (Jarrar and Hamouda, 2024). These resources demonstrate several ways of representing dialectal lexical information. Gahwa follows the same general resource-building tradition, but focuses on standardizing and enriching already existing Emirati evidence rather than collecting a new dictionary from a single source.

Emirati Arabic Resources. Over the past decade, Emirati Arabic has been represented in a growing collection of resources for written text, speech, child language, code-mixed language, sociolinguistic analysis, and automatic speech recognition (Ntelitheos and Idrissi, 2017; Khalifa et al., 2017, 2016, 2018; Al Ali and Aldarmaki, 2024; Alblooki et al., 2025; Alkaabi et al., 2026; Al-Sabbagh, 2026). As summarized in Table 1, these resources have substantially expanded the empirical basis for Emirati Arabic NLP. However, they were developed for different tasks and differ in content, annotation, and accessibility, resulting in fragmented lexical evidence rather than a unified computational lexicon.

3 Gahwa Design and Construction

Lexicon Representation. Each Gahwa record represents a lexical entry using the fields shown in Table 2. The Index is a unique record identifier. The Entry field contains the base word form represented by the record, while Lemma gives the lemma linked to that form. The Type field marks the relation between the two: LEMMA is used when Entry=Lemma, and WORD is used when Entry is a distinct surface or inflected form linked to the lemma in Lemma. Each record also stores the morphological root, core POS (POS) and CAMEL_POS (Khalifa et al., 2018), English Gloss, MSA Lemma equivalent, CAMELBERT_freq or CAMELBERT-Mix frequency (Inoue et al., 2021), and source of the lexical entry, which we present next.

Sources and Extraction. We build Gahwa by extracting candidate records from three sources; Table 3 summarizes their contribution and the subsequent consolidation flow. First, *Gumar Ann* is a ~200K-token annotated subset of the Emirati portion of the *Gumar* (Khalifa et al., 2016, 2018). It supplies entry forms, lemmas, POS labels, CAMEL POS labels, and English glosses. Second, the AE VERB LEXICON, contains 2,648 verb lemmas with roots, perfective and imperfective paradigm forms, and glosses (Khalifa et al., 2017). Finally, the Gulf Wiktionary export contains 525 semi-structured dictionary records. A single Wiktionary record may bundle several forms, POS blocks, or glosses; splitting these combinations produces candidate records whose pairings are then manually checked by two of the co-authors.

Enrichment. After mapping source fields and POS labels to the lexicon representation (Table 2), we fill fields not supplied consistently by the sources. For *Gumar Ann*, we add roots; for Gulf

Field Name	Description	Example
Index	Unique record identifier.	382
Type	LEMMA or WORD.	LEMMA
Entry	Base word form represented by the record.	رَمَسَ <i>ramas</i>
Lemma	Lemma linked to the represented form.	رَمَسَ <i>ramas</i>
Root	Templatic consonantal root or NTWS (non-templatic word stem).	ر م س
POS	Core part-of-speech category.	VERB
CAMEL_POS	Fine-grained part-of-speech morphological category.	VERB.P3MS
English_Gloss	English gloss of the lemma.	speak; speak to someone; talk
MSA_Lemma	Modern Standard Arabic lemma equivalent.	تَكَلَّمَ <i>takal~am</i>
CAMELBERT_Freq	Lemma frequency from CAMELBERT-Mix counts.	2,122
Source	Original source of the lexical entry.	AE_Verb_Lexicon

Table 2: Illustrative Gahwa record for the verb رَمَسَ *ramas* ‘to speak’.

Wiktionary, we add CAMEL_POS. For AE VERB LEXICON, the paradigm projection produces the candidate records reported in Table 3; during enrichment, we add command forms to these projected lemma and word records. For *Gumar Ann*, candidate roots are extracted from the existing Lemma field and checked by two co-authors with Arabic linguistic expertise; records sharing the same Lemma receive the same checked root. MSA lemma equivalents are then added for all three sources. Candidate MSA lemmas are generated using an LLM from the Gahwa Lemma, POS, and English gloss, then checked and corrected by the same co-authors. This process is described further in the *MSA Lemma Generation and Review* paragraph below. Finally, we compute and retain a CAMELBERT-Mix lemma frequency for each Gahwa entry (Inoue et al., 2021).

Preparation and Consolidation. After extraction, we prepare records for comparison by normalizing their English glosses. For each record, we split semicolon-separated gloss lists, trim whitespace, remove punctuation, compare strings case-insensitively, remove duplicate items, and sort the remaining items alphabetically. We use this normalized gloss in all later gloss comparisons.

We consolidate records in two passes, first within each source and then across sources. We remove exact duplicates with the key Entry + Lemma + POS + CAMEL_POS + normalized English gloss. Remaining records are merged only in two cases: (i) when POS and normalized gloss agree and lemmas differ only in diacritization, we keep the more diacritically informative lemma, prioritizing shadda and sukun over other diacritics; and (ii) when lemma

and POS agree and the normalized gloss lists overlap, we keep their normalized union, retaining the fuller list when one list contains the other. All other records remain separate; each merged group or singleton becomes a consolidated entry.

After consolidation, the distinct lexical-entry unit in Gahwa is lemma-POS-gloss (LPG). The lemma gives the lexical head, POS separates category-specific uses, and the normalized English gloss distinguishes meanings that share the same lemma and POS. As shown in Table 3, the process yields 33,687 consolidated records and 10,456 distinct LPG entries.

MSA Lemma Generation and Review. We generate one candidate MSA lemma for each of the 10,456 consolidated lemma-POS-gloss (LPG) entries. The generation process therefore follows a one-to-one mapping from an LPG entry to a candidate MSA lemma. For each LPG entry, the prompt provides the Gulf Arabic lemma (Lemma), POS, and English gloss that constitute the LPG triplet, together with the associated Entry and dialect code (DIA). We use gpt-5-nano-2025-08-07 to return exactly one candidate MSA lemma in Arabic script.

All 10,456 generated LPG-to-MSA-lemma mappings are manually checked and corrected. One co-author performs the primary review, while a second co-author supports the resolution of ambiguous cases. Disagreements are resolved through discussion. Because the two reviews are not conducted independently, we do not report an inter-annotator agreement statistic. The complete prompt and generation settings are provided in Appendix B.

Inventory	Entries
<i>Gumar Ann</i> records	24,607
AE Verb Lexicon records	7,924
Gulf Wiktionary records	1,161
Consolidated records	33,687
Distinct Lemmas (L)	8,890
Distinct Lemma-POS pairs (LP)	9,339
Distinct Lemma-POS-gloss entries (LPG)	10,456

Table 3: Size of the source-specific, consolidated, and derived lexical inventories in Gahwa.

4 Coverage Evaluation

Evaluation data. To our knowledge, no independent Emirati lexical gold standard is available. We therefore use three Emirati datasets as in-domain diagnostics of practical lexical coverage: *GumarAE*, the Emirati subset of *Gumar* (Khalifa et al., 2016),² Mixat (Al Ali and Aldarmaki, 2024), and the Ramsa Corpus (Al-Sabbagh, 2026). For all of these sets, we remove punctuation- or symbol-only items, digit-containing items, foreign-language or web markers, and Arabic strings with attached symbols. The final evaluation corpus sizes are as follows: *GumarAE* (9,052,103 Tokens, and 442,230 Types), Mixat (91,865 Tokens, and 15,636 Types), and Ramsa Corpus (26,489 Tokens, and 7,873 Types).

Analyzers and Metrics. We use three morphological analysis databases distributed with CAMEL TOOLS (Obeid et al., 2020) for Gulf Arabic (GLF), Egyptian Arabic (EGY), and Modern Standard Arabic (MSA).³ ANY denotes their union. A type is in-vocabulary (INV) if any analyzer returns an analysis, and out-of-vocabulary (OOV) otherwise.

Because the CAMEL TOOLS GLF analyzer does not provide English glosses, we evaluate Gahwa at the Lemma-POS (LP) level. POS matching is largely a case-insensitive identity mapping between the two tagsets, e.g. prep=PREP. We use two LP matching conditions: *Exact* compares lemma strings as given, whereas *dediacritized* matching (*Dediac*) strips Arabic diacritics before comparison. In Table 4, LP-match columns use analyzer-covered output as the denominator, while effective coverage uses the full evaluation corpus.

²We use Gumar N-grams as our source for this data (Khalifa et al., 2016): <https://github.com/CAMEL-Lab/Gumar-Ngrams/releases>.

³calima-glf-01 (GLF), calima-egy-r13 (EGY, only analyses marked [CALIMA]), and camel-morph-msa-v1.2.2 (MSA).

Repeated-letter Normalization (RepNorm).

We apply repeated-letter normalization (RepNorm) to all OOV forms by progressively collapsing repeated character sequences. For example, محمود محمود محمود $mHmwwwwwd \rightarrow mHmwwd \rightarrow mHmwd$ ‘Mahmoud’. We first reduce each repeated sequence to at most two characters; if the resulting form remains OOV, we reduce it further to a single character. Results are shown in the last row in Table 4. For *GumarAE*, overall analyzer type coverage increases from 63.3% to 84.4% (+21.1 percentage points), while analyzer token coverage rises from 93.3% to 97.3% (+4.0 percentage points). While the *relative* Gahwa token coverage decreases slightly, effective Gahwa token coverage increases from 86.4% to 89.5% Exact and from 88.8% to 92.2% Dediac.

RepNorm has a smaller effect on the two speech-transcript datasets. Analyzer type coverage increases from 95.6% to 95.9% on Mixat and from 95.4% to 95.8% on Ramsa, while analyzer token coverage remains 98.8% on Mixat and increases from 98.3% to 98.4% on Ramsa. Effective Dediac Gahwa token coverage reaches 94.2% on Mixat and 89.1% on Ramsa. These smaller changes indicate that repeated-letter elongation is less common in the two speech-transcript datasets.

We acknowledge that these matches may not necessarily be the correct analysis in context, but represent a possible analysis nonetheless.

Error Analysis. After RepNorm, 15.6% of *GumarAE* corpus types remain unanalyzed, compared with 4.2% of Ramsa types and 4.1% of Mixat types. Because the *GumarAE* residual inventory is considerably larger, we retain the systematic sample of 1,000 residual types from the earlier analyzer run. For the smaller datasets, we examine all remaining OOV types: 331 Ramsa types and 638 Mixat types. All types included in the three error-analysis sets are manually checked and assigned a single dominant error category by the authors. For Ramsa and Mixat, preliminary LLM-assisted labels were used as a starting point, but every label was subsequently manually verified and corrected using the same categories and criteria applied to *GumarAE*. When multiple phenomena overlap, the category judged to be the primary cause of analyzer failure is selected.

Results are in Table 5. Dialect / lexical gap is the largest residual category, comprising 333 sampled types (33.3%) and 6,784 represented corpus

Analyzer	Analyzer INV		Gahwa Types		Gahwa Tokens		Effective Coverage	
	Types%	Tokens%	Exact%	Dediac%	Exact%	Dediac%	Exact%	Dediac%
GumarAE								
GLF	37.5	83.7	77.7	93.5	84.4	95.4	70.6	79.8
EGY	35.9	79.3	60.8	75.7	76.7	89.5	60.8	71.0
MSA	52.0	87.3	48.8	69.1	63.8	86.8	55.7	75.7
ANY	63.3	93.3	71.3	80.2	92.6	95.2	86.4	88.8
ANY + RepNorm	84.4	97.3	71.3	80.9	92.0	94.8	89.5	92.2
Mixat								
GLF	70.8	89.6	78.9	94.8	83.5	94.0	74.9	84.3
EGY	75.8	91.0	67.1	81.5	75.5	90.1	68.7	82.0
MSA	87.8	95.7	58.6	78.6	65.0	86.7	62.2	82.9
ANY	95.6	98.8	80.7	86.6	92.8	95.3	91.6	94.1
ANY + RepNorm	95.9	98.8	80.6	86.5	92.7	95.3	91.7	94.2
Ramsa Corpus								
GLF	66.6	81.7	78.1	94.2	82.9	94.3	67.7	77.1
EGY	76.8	88.1	64.0	77.0	72.6	86.2	64.0	76.0
MSA	90.7	96.3	54.7	75.3	60.8	83.6	58.6	80.6
ANY	95.4	98.3	76.9	82.6	86.5	90.5	85.0	89.0
ANY + RepNorm	95.8	98.4	76.8	82.6	86.5	90.5	85.1	89.1

Table 4: Coverage and Gahwa lemma-POS agreement across the three Emirati evaluation datasets: **GumarAE** (9,052,103 Tokens, and 442,230 Types), **Mixat** (91,865 Tokens, and 15,636 Types), and **Ramsa Corpus** (26,489 Tokens, and 7,873 Types). ANY is the union of GLF, EGY, and CAMELMorph MSA v1.2.2.

Category	GumarAE		Ramsa Corpus		Mixat	
	Token%	Types%	Token%	Types%	Token%	Types%
Dialect / lexical gap	78.6	33.3	45.9	43.8	51.5	43.1
Orthographic / spelling variation	6.3	21.4	12.2	14.2	8.4	11.6
Fused expression / tokenization	6.8	21.0	4.5	5.1	16.2	13.1
Dialect morphology / clitic gap	4.3	11.6	17.7	19.6	15.3	21.9
Proper name / named entity	1.6	3.3	3.8	2.7	1.7	2.1
Loanword / foreign form	1.3	4.3	11.0	10.2	5.3	6.1
Unclear / ambiguous	0.7	5.1	4.5	4.2	1.3	1.8

Table 5: Residual analyzer-OOV categories after RepNorm. GumarAE is based on a systematic sample of 1,000 residual types (8,623 tokens), while the smaller Ramsa and Mixat datasets include all remaining OOV types: 331 (416 tokens) and 638 (1,070 tokens), respectively.

occurrences (78.7%). Examples include ريو لي *ry-wly* ‘my feet’ and آمبيه *Āmbyh* ‘oh dear!’. Orthographic variation is next in prevalence (21.4% of types, 6.3% of tokens), such as اتصيح *AtSyH* ‘she cries’. Another common category is fused expressions (21% types) such as الحمد لله *AlHmdllh* ‘Thank God’. Dialect / lexical gap is also the largest category in Ramsa and Mixat, comprising 43.8% and 43.1% of residual types, respectively. Morphology / clitic gaps are more prominent in Ramsa and Mixat (19.6% and 21.9% of types) than in the *GumarAE* sample (11.6%). Ramsa also contains a larger proportion of loanwords or foreign forms (10.3% of types). Other lower frequency errors include missing proper names, e.g., حميرا *jmyrA* ‘Jumeirah’.

5 Conclusion and Future Work

We presented Gahwa, the broadest **open-source** computational lexicon for Emirati Arabic to date, consolidating lexical information from diverse sources. Evaluation on *GumarAE*, *Mixat*, and the *Ramsa Corpus* shows that Gahwa achieves 92.2%, 94.2%, and 89.1% effective dediacritized coverage of running tokens; residual OOV analysis identifies the main remaining gaps: dialect vocabulary, orthographic variation, fused expressions, and morphology or clitic gaps.

Future work will expand Gahwa using the residual OOV analysis and improve Emirati morphological analyzers to better support its lexical inventory. We envision Gahwa as a component of the infrastructure for Emirati Arabic NLP.

Limitations

Our coverage evaluation is diagnostic and measures analyzability rather than the correctness of the selected lemma/lemmaPOS analysis. One resource used to construct Gahwa, *Gumar Ann*, is derived from material related to *GumarAE*, although it represents only about 1.5% of its size. Coverage estimates also depend on the available morphological analyzers, whose own lexical limitations may underestimate the true coverage of Gahwa. Finally, both Gahwa and its evaluation are constrained by the currently available Emirati resources. Although we evaluate on *GumarAE*, Mixat, and the Ramsa Speech Corpus, these datasets may still underrepresent lexical variation across domains and varieties of Emirati Arabic.

Ethics Statement

Gahwa is constructed from existing Emirati and Gulf Arabic lexical resources used in accordance with their respective licenses and terms of use. All manual inspection, annotation, and error analysis were performed by the authors.

Like many NLP resources, Gahwa could be misused for applications such as language profiling. We encourage its use only for ethical research and development, in accordance with applicable laws, institutional policies, and professional ethical guidelines.

AI writing assistance was used only for language-level editing of the manuscript, in accordance with the ACL Policy on Publication Ethics; all resource construction, analyses, experiments, and reported results were designed, produced, and verified by the authors.

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A License

We list below the licenses of the data and tools used in this work, all of which are employed in accordance with their intended use.

- Gumar Corpus N-grams (Khalifa et al., 2016) CC BY 3.0. <https://github.com/CAMeL-Lab/Gumar-Ngrams/releases>
- AE Verb Lexicon (Khalifa et al., 2017): CC BY 3.0.
- Gumar Annotated Corpus (Khalifa et al., 2018): NYU Abu Dhabi Non-commercial, research-only license.
- Morphological analyzers provided by CAMeL Tools (Obeid et al., 2020) and CAMeL-BERT (Inoue et al., 2021): MIT License.
- Wiktionary (Krumbiegel, 2023): CC BY-SA 4.0.
- Camel Morph MSA (camel-morph-msa-v1.2.2): CC BY 4.0.

B MSA Lemma Generation and Review

We use gpt-5-nano-2025-08-07 to generate one candidate MSA lemma for each of the 10,456 lemma-POS-gloss (LPG) entries. The generation process is therefore a one-to-one mapping: The LPG triplet consists of the Gahwa Lemma, POS, and English gloss. The associated Entry and dialect code (DIA) are also included in the request. All 10,456 generated MSA lemmas are manually checked and corrected by the co-authors.

No temperature, top-p, seed, or maximum output-token setting is explicitly specified; the API defaults are used.

System Prompt

Model used for the final MSA lemmas: gpt-5-nano-2025-08-07

Number of LLM-generated MSA lemmas/LLM checked Lemmas: 10,456 LPG entries

Full system prompt:

You are an expert Arabic linguist specializing in Gulf-MSA lexical mapping.

You will receive several columns taken from a Gulf Arabic lexicon.

Use ALL of the following columns when determining the correct Modern Standard Arabic (MSA) lemma:

1. Lex : the Gulf Arabic lemma (dialectal base form), typically fully diacritized.
 - This is the primary and most reliable piece of information.
 - Treat it as the strongest anchor for identifying the correct MSA lemma.
2. POS : the part of speech.
 - Use this to determine the correct MSA lemma form:
 - VERB : return the MSA lemma in its dictionary form: the fully diacritized 3rd person masculine singular perfective
 - NOUN : return the fully diacritized singular indefinite nominative form
 - ADJ : return the fully diacritized masculine singular indefinite form
3. Entry : the surface form as it appears in corpus data.
 - Use this only as a secondary hint if needed.
4. DIA : dialect code (e.g., AE).
 - Use this to understand the intended Gulf variety and resolve dialect-specific meanings.
5. English Gloss : a concatenated or noisy gloss.
 - Use it ONLY as a weak sense-disambiguation hint.
 - If the gloss conflicts with Lex or POS, ignore the gloss completely.

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DEFINITION OF AN MSA LEMMA
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A lemma is the canonical, dictionary citation form of a word. It must follow these rules:

VERB Lemmas

- Return the fully diacritized 3rd person masculine singular perfective form.
- Do not include any mood diacritic.
- No attached particles or clitics of any kind (such as لا، لم، س، ف، و or pronoun suffixes).
- Use the dialect lemma's diacritization as a hint but correct it to match the proper MSA lemma.

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GENERAL RULES FOR ALL LEMMAS
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- The lemma must be:
 - Fully diacritized
 - A single canonical stem
 - Using underscores "_" instead of spaces
 - Without attached particles, clitics, conjunctions, prepositions, or determiners
 - Without case endings or verb mood endings
- The lemma must not:
 - Contain case endings
 - Contain verb mood endings
 - Contain the determiner al-
 - Contain plural or dual suffixes (for nouns/adjectives)
 - Be undiacritized or partially diacritized

Your task:

- Map the Gulf Arabic lemma to exactly ONE correct MSA lemma.
- The result must be:
 - a single MSA lemma

- in Arabic script
- with diacritics
- with correct spelling
- using underscores `\_` instead of spaces

Output only valid JSON in exactly this format:

```
{"msa_lemma": "<single MSA lemma with underscores>"}
```

Do NOT output multiple lemmas.

Do NOT output semicolons.

Do NOT output anything else: no explanations, no comments, no transliterations.